PeoplePay360 — Complete Database Schema

PostgreSQL blueprint for the integrated HR → Attendance/Leave → Payroll Engine → Explainable Payslip workflow.

Architecture

One PostgreSQL database, logically separated into identity, organization, HR, attendance, leave, payroll configuration, payroll execution, and audit domains. The Payroll Engine is an application layer. Finalized payroll is preserved through payslip lines, inputs, calculation traces and snapshots.

> **Core flow:** Employee → Contract → Schedule → Attendance/Leave → Salary Structure + Versioned Rules → Payrun → Payslip → Calculation Trace → Validation → Payment/Delivery.

1. Database Modules

Schema	Tables
identity	users, roles, user_roles
organization	departments, job_positions
hr	employees, employee_bank_accounts, working_schedules, working_schedule_days, contracts
attendance	attendance_records
leave_management	time_off_types, leave_allocations, leave_requests
payroll_config	salary_structures, salary_rules, salary_rule_versions, salary_structure_rules
payroll	payruns, payrun_employees, payslips, payslip_lines, payslip_inputs, payslip_worked_days, payslip_calculation_trace, payroll_warning
audit	audit_logs

2. Complete Table Definitions

identity.users

Column	Type	Constraints / FK	Purpose
id	UUID	PK, gen_random_uuid()	Account ID
email	VARCHAR(255)	NOT NULL, UNIQUE	Login email
password_hash	TEXT	NOT NULL	Password hash
status	user_status	NOT NULL	ACTIVE/INACTIVE/LOCKED
last_login_at	TIMESTAMPTZ		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	

identity.roles

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
name	VARCHAR(100)	NOT NULL, UNIQUE	EMPLOYEE/HR_MANAGER/HR_PAYROLL_USER/HR_PAYROLL_MANAGER/ADMIN
description	TEXT		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	

identity.user_roles

Column	Type	Constraints / FK	Purpose
user_id	UUID	PK/FK → identity.users.id	
role_id	UUID	PK/FK → identity.roles.id	
assigned_at	TIMESTAMPTZ	DEFAULT NOW()	

organization.departments

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
name	VARCHAR(150)	NOT NULL	
code	VARCHAR(50)	NOT NULL, UNIQUE	
manager_employee_id	UUID	FK → hr.employees.id, NULL	Add FK after employees
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

organization.job_positions

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
name	VARCHAR(150)	NOT NULL	
description	TEXT		
department_id	UUID	NOT NULL, FK → departments.id	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

hr.employees

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
user_id	UUID	UNIQUE, FK → identity.users.id, NULL	Optional login
employee_code	VARCHAR(50)	NOT NULL, UNIQUE	
first_name	VARCHAR(100)	NOT NULL	
last_name	VARCHAR(100)	NOT NULL	
email	VARCHAR(255)	NOT NULL, UNIQUE	
phone	VARCHAR(30)		
date_of_birth	DATE		

Column	Type	Constraints / FK	Purpose
date_of_joining	DATE	NOT NULL	
date_of_leaving	DATE		
department_id	UUID	FK → organization.departments.id	
job_position_id	UUID	FK → organization.job_positions.id	
manager_id	UUID	FK → hr.employees.id	
employment_type	employment_type	NOT NULL	FULL_TIME/PART_TIME/CONTRACT/INTERN
status	employee_status	NOT NULL DEFAULT ACTIVE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

hr.employee_bank_accounts

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
employee_id	UUID	NOT NULL, FK → employees.id	
account_holder_name	VARCHAR(200)	NOT NULL	
bank_name	VARCHAR(150)	NOT NULL	
account_number	VARCHAR(100)	NOT NULL	Encrypt/mask in UI
ifsc_code	VARCHAR(20)		
is_primary	BOOLEAN	DEFAULT FALSE	
is_verified	BOOLEAN	DEFAULT FALSE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

hr.working_schedules

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
name	VARCHAR(150)	NOT NULL	
schedule_type	VARCHAR(50)	NOT NULL	
timezone	VARCHAR(80)	NOT NULL DEFAULT 'Asia/Kolkata'	
weekly_hours	NUMERIC(6,2)	NOT NULL DEFAULT 0	Calculate from schedule days
is_active	BOOLEAN	DEFAULT TRUE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

hr.working_schedule_days

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
schedule_id	UUID	NOT NULL, FK → working_schedules.id	
day_of_week	SMALLINT	NOT NULL, CHECK 0..6	
is_working_day	BOOLEAN	DEFAULT TRUE	
start_time	TIME		
end_time	TIME		
break_minutes	INTEGER	DEFAULT 0, CHECK >=0	
expected_hours	NUMERIC(5,2)	DEFAULT 0	Calculated

hr.contracts

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
employee_id	UUID	NOT NULL, FK → employees.id	
contract_number	VARCHAR(80)	NOT NULL, UNIQUE	
start_date	DATE	NOT NULL	
end_date	DATE		
status	contract_status	NOT NULL DEFAULT DRAFT	
wage	NUMERIC(14,2)	NOT NULL, CHECK >=0	

Column	Type	Constraints / FK	Purpose
currency	CHAR(3)	NOT NULL DEFAULT 'INR'	
department_id	UUID	FK → departments.id	
job_position_id	UUID	FK → job_positions.id	
schedule_id	UUID	FK → working_schedules.id	
salary_structure_id	UUID	FK → payroll_config.salary_structures.id	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

attendance.attendance_records

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
employee_id	UUID	NOT NULL, FK → employees.id	
work_date	DATE	NOT NULL	
check_in	TIMESTAMPTZ		
check_out	TIMESTAMPTZ		
worked_minutes	INTEGER	DEFAULT 0, CHECK >=0	Calculated/validated
overtime_minutes	INTEGER	DEFAULT 0, CHECK >=0	
status	VARCHAR(40)	NOT NULL	PRESENT/LATE/ABSENT/HALF_DAY/OVERTIME/MISSING_CHECKOUT/CORRECTED
source	VARCHAR(40)	DEFAULT 'MANUAL'	
is_manual_edit	BOOLEAN	DEFAULT FALSE	
corrected_by	UUID	FK → identity.users.id	
correction_reason	TEXT		
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

leave_management.time_off_types

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
name	VARCHAR(120)	NOT NULL	
code	VARCHAR(50)	NOT NULL, UNIQUE	
unit	VARCHAR(20)	NOT NULL	DAYS/HOURS
requires_allocation	BOOLEAN	DEFAULT TRUE	
requires_approval	BOOLEAN	DEFAULT TRUE	
payroll_behavior	VARCHAR(30)	NOT NULL	PAID/UNPAID/PARTIALLY_PAID/EXCLUDED
is_active	BOOLEAN	DEFAULT TRUE	

leave_management.leave_allocations

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
employee_id	UUID	NOT NULL, FK → employees.id	
time_off_type_id	UUID	NOT NULL, FK → time_off_types.id	
valid_from	DATE	NOT NULL	
valid_to	DATE	NOT NULL	
allocated_units	NUMERIC(10,2)	CHECK >0	
used_units	NUMERIC(10,2)	DEFAULT 0	Cached/derived
remaining_units	NUMERIC(10,2)	DEFAULT 0	Cached/derived
status	VARCHAR(30)	DEFAULT 'DRAFT'	
approved_by	UUID	FK → identity.users.id	
approved_at	TIMESTAMPTZ		

leave_management.leave_requests

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
employee_id	UUID	NOT NULL, FK → employees.id	
time_off_type_id	UUID	NOT NULL, FK → time_off_types.id	
allocation_id	UUID	FK → leave_allocations.id	
start_date	DATE	NOT NULL	
end_date	DATE	NOT NULL	
requested_units	NUMERIC(10,2)	CHECK >0	
reason	TEXT		
status	VARCHAR(30)	DEFAULT 'PENDING'	DRAFT/PENDING/APPROVED/REJECTED/CANCELLED
requested_at	TIMESTAMPTZ	DEFAULT NOW()	
approved_by	UUID	FK → identity.users.id	
approved_at	TIMESTAMPTZ		
rejection_reason	TEXT		

payroll_config.salary_structures

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
name	VARCHAR(150)	NOT NULL	
code	VARCHAR(60)	NOT NULL, UNIQUE	
description	TEXT		
is_active	BOOLEAN	DEFAULT TRUE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

payroll_config.salary_rules

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
name	VARCHAR(150)	NOT NULL	
code	VARCHAR(60)	NOT NULL, UNIQUE	
category	VARCHAR(30)	NOT NULL	BASIC/ALLOWANCE/GROSS/DEDUCTION/CONTRIBUTION/NET
description	TEXT		
is_active	BOOLEAN	DEFAULT TRUE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

payroll_config.salary_rule_versions

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
salary_rule_id	UUID	NOT NULL, FK → salary_rules.id	
version_number	INTEGER	NOT NULL, CHECK >0	
calculation_type	VARCHAR(30)	NOT NULL	FIXED/PERCENTAGE/FORMULA
fixed_amount	NUMERIC(14,2)		
percentage	NUMERIC(8,4)		
base_code	VARCHAR(60)		Controlled reference, e.g. BASIC
formula_expression	TEXT		Safe expression language; never eval arbitrary code
condition_expression	TEXT		Optional safe condition
effective_from	DATE	NOT NULL	
effective_to	DATE		
is_active	BOOLEAN	DEFAULT TRUE	
created_by	UUID	FK → identity.users.id	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

payroll_config.salary_structure_rules

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
salary_structure_id	UUID	NOT NULL, FK → salary_structures.id	
salary_rule_id	UUID	NOT NULL, FK → salary_rules.id	
rule_version_id	UUID	FK → salary_rule_versions.id	Pin exact version when needed
sequence	INTEGER	NOT NULL, CHECK >0	Execution order
is_active	BOOLEAN	DEFAULT TRUE	

payroll.payruns

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
run_number	VARCHAR(80)	NOT NULL, UNIQUE	
name	VARCHAR(150)	NOT NULL	
period_start	DATE	NOT NULL	
period_end	DATE	NOT NULL	
salary_structure_id	UUID	NOT NULL, FK → salary_structures.id	
status	VARCHAR(30)	DEFAULT 'DRAFT'	DRAFT/COMPUTING/COMPUTED/VALIDATING/VALIDATED/PAID/CANCELLED
created_by	UUID	NOT NULL, FK → identity.users.id	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
computed_at	TIMESTAMPTZ		
validated_at	TIMESTAMPTZ		
paid_at	TIMESTAMPTZ		

payroll.payrun_employees

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
payrun_id	UUID	NOT NULL, FK → payruns.id	
employee_id	UUID	NOT NULL, FK → employees.id	
contract_id	UUID	NOT NULL, FK → contracts.id	Contract resolved for period
status	VARCHAR(30)	DEFAULT 'SELECTED'	SELECTED/PROCESSING/COMPUTED/ERROR
included_at	TIMESTAMPTZ	DEFAULT NOW()	

payroll.payslips

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
payrun_id	UUID	NOT NULL, FK → payruns.id	
employee_id	UUID	NOT NULL, FK → employees.id	
contract_id	UUID	NOT NULL, FK → contracts.id	
salary_structure_id	UUID	NOT NULL, FK → salary_structures.id	
period_start	DATE	NOT NULL	
period_end	DATE	NOT NULL	
worked_days	NUMERIC(8,2)	DEFAULT 0	
paid_leave_days	NUMERIC(8,2)	DEFAULT 0	
unpaid_leave_days	NUMERIC(8,2)	DEFAULT 0	
overtime_hours	NUMERIC(8,2)	DEFAULT 0	
gross_salary	NUMERIC(14,2)	DEFAULT 0	
total_deductions	NUMERIC(14,2)	DEFAULT 0	
total_contributions	NUMERIC(14,2)	DEFAULT 0	
net_salary	NUMERIC(14,2)	DEFAULT 0	
status	VARCHAR(30)	DEFAULT 'DRAFT'	DRAFT/COMPUTED/VALIDATED/PAID/CANCELLED
currency	CHAR(3)	DEFAULT 'INR'	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
computed_at	TIMESTAMPTZ		

Column	Type	Constraints / FK	Purpose
validated_at	TIMESTAMPTZ		

payroll.payslip_lines

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
payslip_id	UUID	NOT NULL, FK → payslips.id	
salary_rule_id	UUID	NOT NULL, FK → salary_rules.id	
rule_version_id	UUID	FK → salary_rule_versions.id	
sequence	INTEGER	NOT NULL	
code	VARCHAR(60)	NOT NULL	
name	VARCHAR(150)	NOT NULL	
category	VARCHAR(30)	NOT NULL	
base_amount	NUMERIC(14,2)	DEFAULT 0	
percentage	NUMERIC(8,4)		
computed_amount	NUMERIC(14,2)	DEFAULT 0	
quantity	NUMERIC(12,4)		
is_earning	BOOLEAN	DEFAULT FALSE	
is_deduction	BOOLEAN	DEFAULT FALSE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

payroll.payslip_inputs

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
payslip_id	UUID	NOT NULL, FK → payslips.id	
input_code	VARCHAR(60)	NOT NULL	
input_name	VARCHAR(150)	NOT NULL	
value	NUMERIC(14,4)	NOT NULL	
source_type	VARCHAR(50)	NOT NULL	CONTRACT/ATTENDANCE/LEAVE/SCHEDULE/MANUAL
source_id	UUID		
created_at	TIMESTAMPTZ	DEFAULT NOW()	

payroll.payslip_worked_days

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
payslip_id	UUID	NOT NULL, FK → payslips.id	
work_date	DATE	NOT NULL	
code	VARCHAR(50)	NOT NULL	WORKED/PAID_LEAVE/UNPAID_LEAVE/ABSENT/OVERTIME
scheduled_hours	NUMERIC(6,2)	DEFAULT 0	
worked_hours	NUMERIC(6,2)	DEFAULT 0	
paid_hours	NUMERIC(6,2)	DEFAULT 0	
unpaid_hours	NUMERIC(6,2)	DEFAULT 0	
status	VARCHAR(30)	NOT NULL	
source_attendance_id	UUID	FK → attendance_records.id	
source_leave_id	UUID	FK → leave_requests.id	

payroll.payslip_calculation_trace

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
payslip_id	UUID	NOT NULL, FK → payslips.id	
payslip_line_id	UUID	FK → payslip_lines.id	
step_number	INTEGER	NOT NULL	
rule_code	VARCHAR(60)	NOT NULL	
input_snapshot	JSONB	NOT NULL DEFAULT '{}'	

Column	Type	Constraints / FK	Purpose
base_code	VARCHAR(60)		
base_value	NUMERIC(14,4)		
expression	TEXT		Human-readable safe expression
result	NUMERIC(14,4)	NOT NULL	
explanation	TEXT		
created_at	TIMESTAMPTZ	DEFAULT NOW()	

payroll.payroll_warnings

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
payrun_id	UUID	NOT NULL, FK → payruns.id	
payslip_id	UUID	FK → payslips.id	
employee_id	UUID	FK → employees.id	
warning_code	VARCHAR(80)	NOT NULL	
severity	VARCHAR(20)	NOT NULL	INFO/WARNING/ERROR/BLOCKER
message	TEXT	NOT NULL	
is_resolved	BOOLEAN	DEFAULT FALSE	
resolved_by	UUID	FK → identity.users.id	
resolved_at	TIMESTAMPTZ		

payroll.payslip_snapshots

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
payslip_id	UUID	NOT NULL, UNIQUE, FK → payslips.id	
employee_snapshot	JSONB	NOT NULL	
contract_snapshot	JSONB	NOT NULL	
structure_snapshot	JSONB	NOT NULL	
rules_snapshot	JSONB	NOT NULL	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

payroll.payments

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
payslip_id	UUID	NOT NULL, FK → payslips.id	
employee_id	UUID	NOT NULL, FK → employees.id	
amount	NUMERIC(14,2)	CHECK >=0	
currency	CHAR(3)	DEFAULT 'INR'	
payment_method	VARCHAR(40)	NOT NULL	BANK_TRANSFER/OTHER
transaction_reference	VARCHAR(150)	UNIQUE	
status	VARCHAR(30)	DEFAULT 'PENDING'	PENDING/PROCESSING/SUCCESS/FAILED
initiated_at	TIMESTAMPTZ	DEFAULT NOW()	
completed_at	TIMESTAMPTZ		
failure_reason	TEXT		

payroll.payslip_deliveries

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
payslip_id	UUID	NOT NULL, FK → payslips.id	
delivery_type	VARCHAR(30)	NOT NULL	EMAIL/PDF
recipient	VARCHAR(255)	NOT NULL	
status	VARCHAR(30)	DEFAULT 'PENDING'	PENDING/SENT/FAILED
sent_at	TIMESTAMPTZ		
failure_reason	TEXT		

audit.audit_logs

Column	Type	Constraints / FK	Purpose
id	UUID	PK	
user_id	UUID	FK → identity.users.id	
entity_type	VARCHAR(100)	NOT NULL	
entity_id	UUID	NOT NULL	
action	VARCHAR(50)	NOT NULL	CREATE/UPDATE/DELETE/APPROVE/COMPUTE/VALIDATE/PAY
old_values	JSONB		
new_values	JSONB		
ip_address	INET		
created_at	TIMESTAMPTZ	DEFAULT NOW()	

3. PostgreSQL Types

```
CREATE EXTENSION IF NOT EXISTS pgcrypto; CREATE EXTENSION IF NOT EXISTS btree_gist; CREATE TYPE identity.user_status AS ENUM (
'ACTIVE','INACTIVE','LOCKED' ); CREATE TYPE hr.employee_status AS ENUM ( 'ACTIVE','INACTIVE','TERMINATED' ); CREATE TYPE hr.employment_type
AS ENUM ( 'FULL_TIME','PART_TIME','CONTRACT','INTERN' ); CREATE TYPE hr.contract_status AS ENUM ( 'DRAFT','ACTIVE','EXPIRED','TERMINATED' );
```

4. Critical Constraints & Indexes

```
-- Contract lookup during payroll CREATE INDEX contracts_employee_dates_idx ON hr.contracts(employee_id, start_date, end_date); --
Attendance retrieval by employee and payroll period CREATE INDEX attendance_employee_date_idx ON attendance.attendance_records(employee_id,
work_date); -- Leave retrieval CREATE INDEX leave_requests_employee_dates_idx ON leave_management.leave_requests(employee_id, start_date,
end_date, status); -- Payroll period/status CREATE INDEX payruns_period_status_idx ON payroll.payruns(period_start, period_end, status); --
Payslip history CREATE INDEX payslips_employee_period_idx ON payroll.payslips(employee_id, period_start, period_end); CREATE INDEX
payslip_lines_order_idx ON payroll.payslip_lines(payslip_id, sequence); CREATE INDEX calculation_trace_order_idx ON
payroll.payslip_calculation_trace(payslip_id, step_number); CREATE UNIQUE INDEX payrun_employee_unique_idx ON
payroll.payrun_employees(payrun_id, employee_id); CREATE UNIQUE INDEX structure_rule_unique_idx ON payroll_config.salary_structure_rules(
salary_structure_id, salary_rule_id );
```

5. Contract Integrity

```
-- Recommended PostgreSQL protection against overlapping -- active/draft contracts for the same employee. ALTER TABLE hr.contracts ADD
CONSTRAINT contracts_no_overlap EXCLUDE USING gist ( employee_id WITH =, daterange( start_date, COALESCE(end_date + 1, '9999-12-31'::date),
'[)' ) WITH && ) WHERE (status IN ('DRAFT','ACTIVE'));
```

6. Payroll Engine Data Contract

```
PayrollContext { employee, period, contract, schedule, attendanceInputs, leaveInputs, salaryStructure, salaryRuleVersions } PayrollResult {
inputs, workedDays, payslipLines, gross, deductions, contributions, net, warnings, calculationTrace }
```

7. Payrun State Machine

DRAFT → COMPUTING → COMPUTED → VALIDATING → VALIDATED → PAID

Backend owns state transitions. Paid payroll should not be silently mutated. Corrections should create a reversal/adjustment or a new run.

8. Example Salary Rules

```
10 BASIC = CONTRACT_WAGE 20 HRA = BASIC × 40% 30 TRANSPORT = ■3,000 40 GROSS = BASIC + HRA + TRANSPORT 50 PF = BASIC × 12% 60 PT = ■200 100
NET = GROSS - PF - PT For contract wage ■60,000: BASIC = ■60,000 HRA = ■24,000 TRANSPORT = ■3,000 GROSS = ■87,000 PF = ■7,200 PT = ■200 NET
= ■79,600
```

9. Recommended Implementation Order

Phase	Build
1	Database, extensions, enums, identity, organization
2	Employees, schedules, contracts
3	Attendance and Leave
4	Salary structures, rules, rule versions and sequencing
5	Payruns and employee selection
6	Payroll Context + Rule Engine
7	Payslips, lines, inputs, worked days and calculation trace
8	Warnings, validation, snapshots, payments and delivery
9	Audit, indexes, constraints, idempotency and transaction tests
10	Realistic seed data + complete end-to-end payroll test

Design note: This blueprint is grounded in the supplied PeoplePay360 requirements. Versioned salary rules, calculation traces, snapshots, idempotency and historical immutability are engineering enhancements intended to make the payroll engine auditable and robust.